

I have an ulcer



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Dermatology: a neglected specialty

- Often consider to be non-life threatening
- Often consider to be minor problem
- not well taught in institution



Why is Dermatology important?

- 80% will have skin diseases sometimes in their life
- Dermatology accounts for 15% of GP's consultation
- Associated with significant degree of morbidity and loss in work ability
- Cutaneous manifestation of systemic disease

Undergraduate teaching





Core

- Skin Basic
- Cutaneous manifestation of systemic disease
- Life threatening dermatoses
 - (a) Blistering disease
 - (b) Erythroderma
- Common dermatoses
- Treatment in dermatology



Format

- 2 slides sessions (PBL), 1 out-patients, 1 teaching round per week
- Teaching clinic
- lectures



Case 1

- A 35 yrs old businessman notice painful rash in his penile shaft that start off as vesicles and then develop into ulcers
- he goes to China often but deny exposure to sexual worker. He does have a girlfriend who resides in China
- Clinical examination





Questions

- What are the possibilities?
- Any question you want to ask?
- How will you investigate and manage?



Penile ulcer DDx

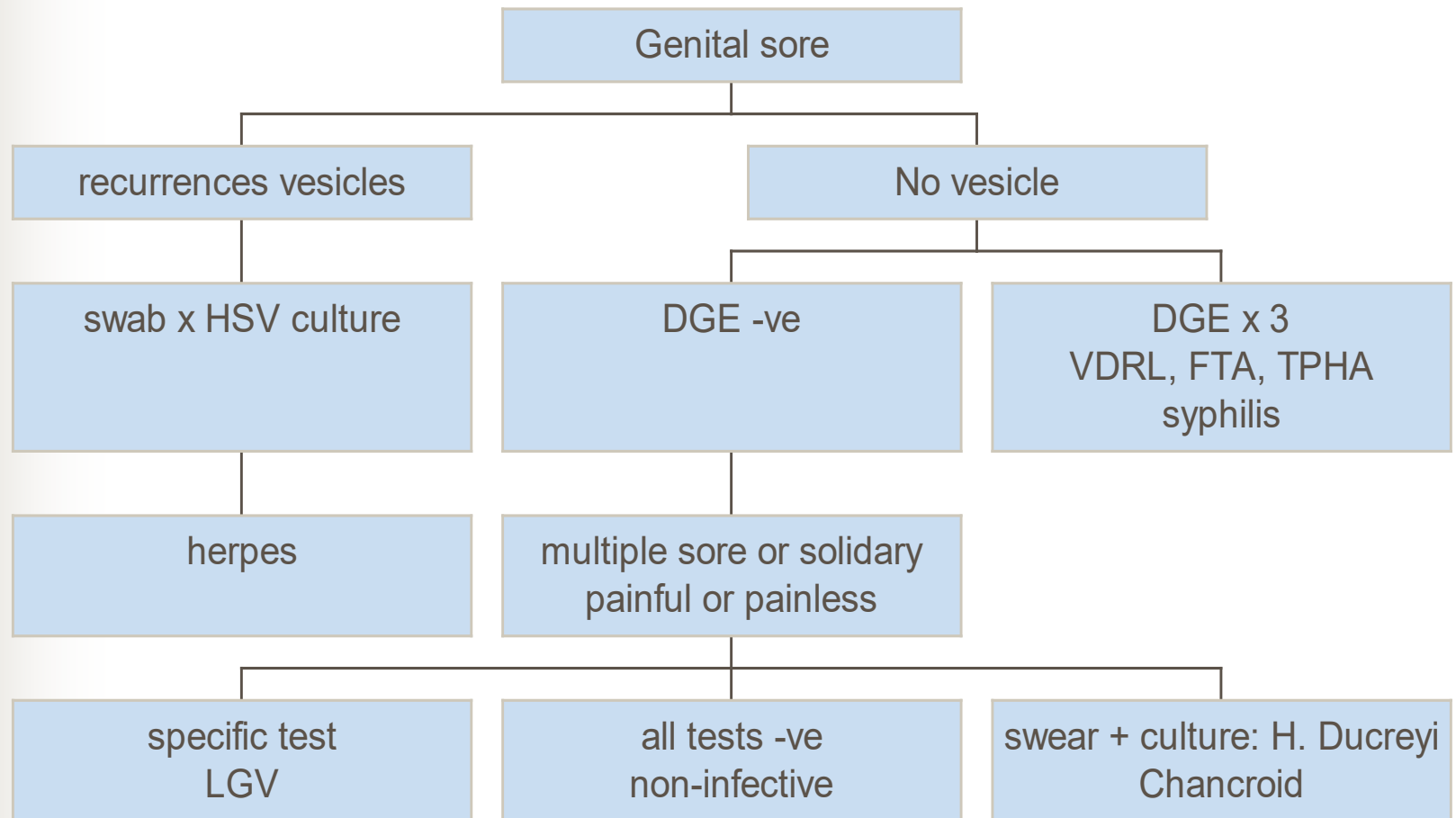
Infective

- Syphilis
- Herpes
- Chancroid
- Lymphogranuloma venereum

Non-infective

- Malignancy
- Trauma
- Fixed drug eruption
- Behcet's syndrome
- Erythema multiforme

Penile ulcer, Investigation





Herpes

- Commonest cause of genital ulcer in HK
- Due to HSV 1(10%) or 2(90%)
- IP: 2 to 5 days
- Recurrent painful ulcers in genitalia preceded by vesicles
- Primary attack particularly painful with tender inguinal lymphadenopathy
- Primary attack can last about three weeks



Recurrent herpes infection

- HSV remain as latent infection in sensory and autonomic ganglion
- Reactivation gives rise to recurrent local infection
- Recurrent: local, less symptomatic, shorter
- Activate by trauma, menstruation, stress



Complication

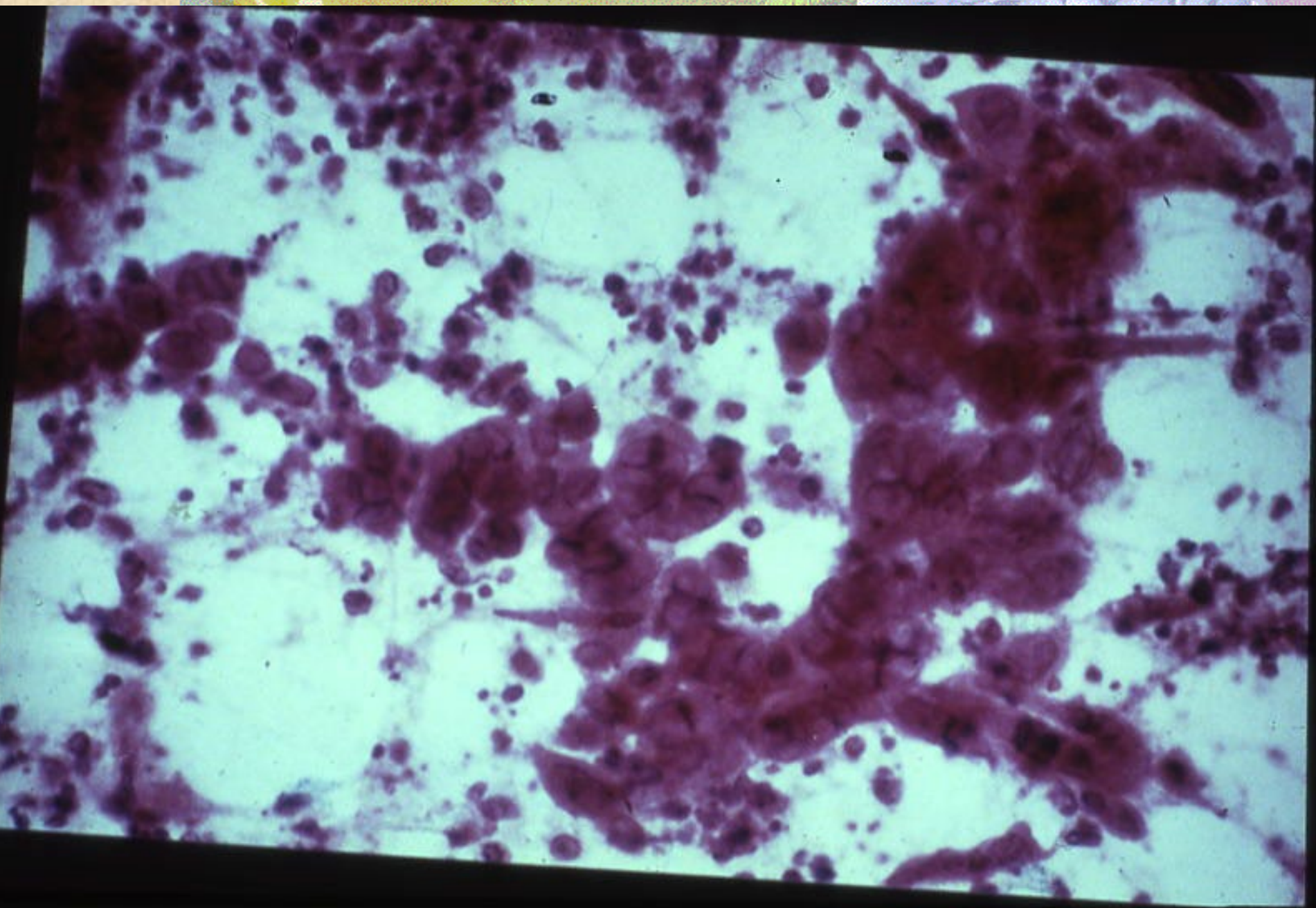
- Cutaneous dissemination
- Encephalitis
- Sacral radiculomyelopathy
- Urinary retention
- Bacterial superinfection
- Erythema multiforme
- Neonatal herpes

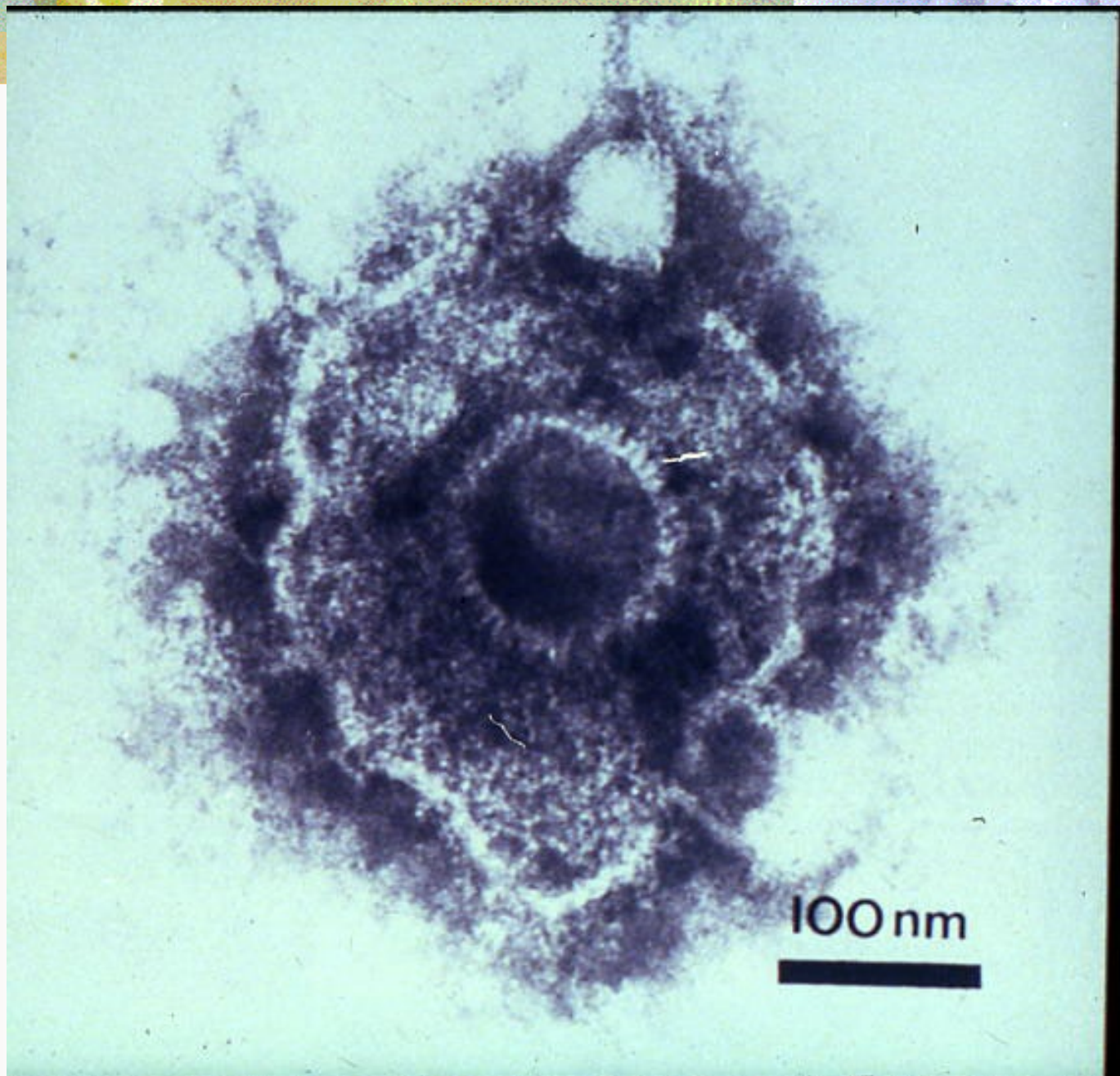




Investigations

- Viral antigen detection
- Viral culture
- Immunoassay (to detect viral antibody)
- EM
- PCR
- Serology (inaccurate)







Management

- Symptomatic control
- Anti-viral agents
- Topical antibiotics to reduce superimposed infection
- Contact tracing
- Screen for other STD



Syphilis

- STD
- *Treponema pallidum*
- IP: 9 to 90 days
- Congenital or acquired
- Primary/secondary/tertiary
- Early/latent



Primary syphilis:chancre

- Solitary, painless ulcer with indurated base
- At site of trauma during sexual intercourse
- Inguinal lymphadenopathy: non-tender, discrete, rubbery







Case 2

- 55 years old lady with ulcer of the legs for several years
- Examination revealed:
- What other questions do you want to ask?
- What are the possibilities?





DDX of leg ulcers

- Venous
- Arterial
- Pyoderma gangrenosum
- Dysglobulinaemias
- Sickle cell and thalassaemia
- Basal cell carcinoma
- Squamous cell carcinoma
- Panniculitis
- Vasculitis
- pressure



Varicose eczema and ulceration

- Venous hypertension causing distension and elongation of the capillary loops
- Fibrinogen leaks into the tissue and form cuffs in the capillary
- Prevent diffusion of O₂ and nutrients to the skin
- Local ischaemia caused ulceration



Clinical features

- Eczema
- Atrophie blanche
- Oedema
- Pigmentation
- Ulceration
- Venous flare
- varicosities



Ulceration

- After minor injury
- Often painless
- Well defined border
- Shallow ulcer with yellowish base
- Medial and lateral malleoli



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Treatment

- Rest and leg elevation
- Compression dressing
- Exercise
- Diet to reduce weight
- Treatment of eczema
- Antibiotic if superimposed infection
- Cleansing agent
- surgical



Arterial leg ulcers

- Arterial insufficiency leading to skin ulceration
- Risk factors include smoking, diabetes, HT
- Painful ulcer worse at light
- Other symptoms of ischaemia common:
Intermittent claudication, coldness,
numbness









Signs

- Skin dry and scaly
- Hair sparse or absence
- Toenail thickened
- Deep punch-out ulcer
- Absent pulse
- gangrene



Arterial vs venous

Arterial

- Painful
- Toes, heels, shins
- Pulse reduce or absence
- Dry, scaly, atrophic, shiny skin
- Hair absence
- Nail dystrophic
- Pallor, cyanosis

Venous

- Painless
- Malleoli
- Pulse normal
- Eczema
- Pigmentation
- Hair normal
- Normal nail
- No pallor, cyanosis



Management

- Doppler US
- Arteriographic assessment
- Control of risk factors
- Surgical assessment
- Elastic stocking contraindicated



Pyoderma gangrenosum

- Painful nodule or pustule break down to from progressive enlarge ulcer
- Single or multiple
- Irregular, purplish border
- Known association with systemic diseases





Association with systemic disease

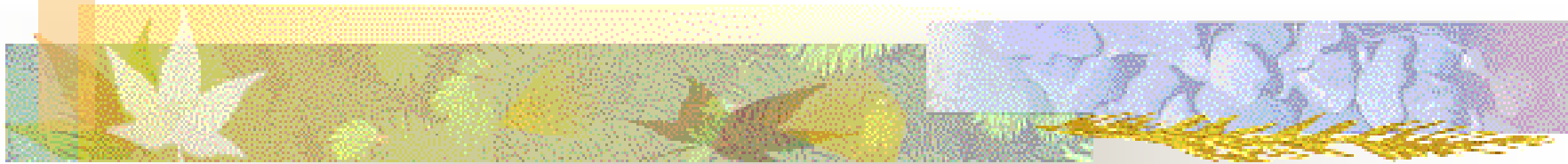
- Inflammatory bowel diseases
- Connective tissue diseases
- Arthritis
- Gammopathy
- haematological malignancy
- Other(Bechet's disease)



Investigation and management

- Exclude infection by skin swab and tissue culture
- Skin biopsy
- Pathergy test
- Investigate for associated conditions
- Treat by the use of systemic immunosuppressive and also underlying disease

I am itchy but no rash

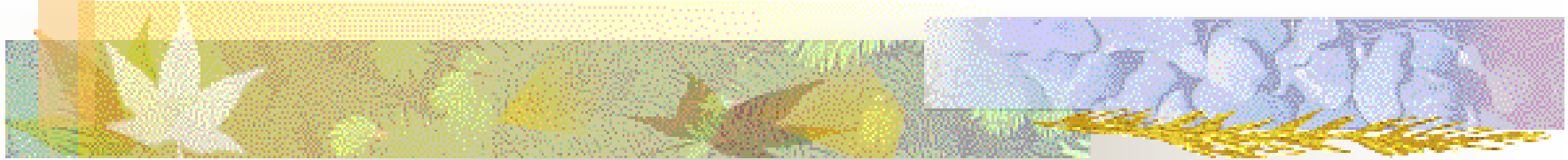




Case 3

- 65 yrs old elderly home resident with itchness affecting her whole body but spare the face
- The symptom is worse at night
- Clinical examination

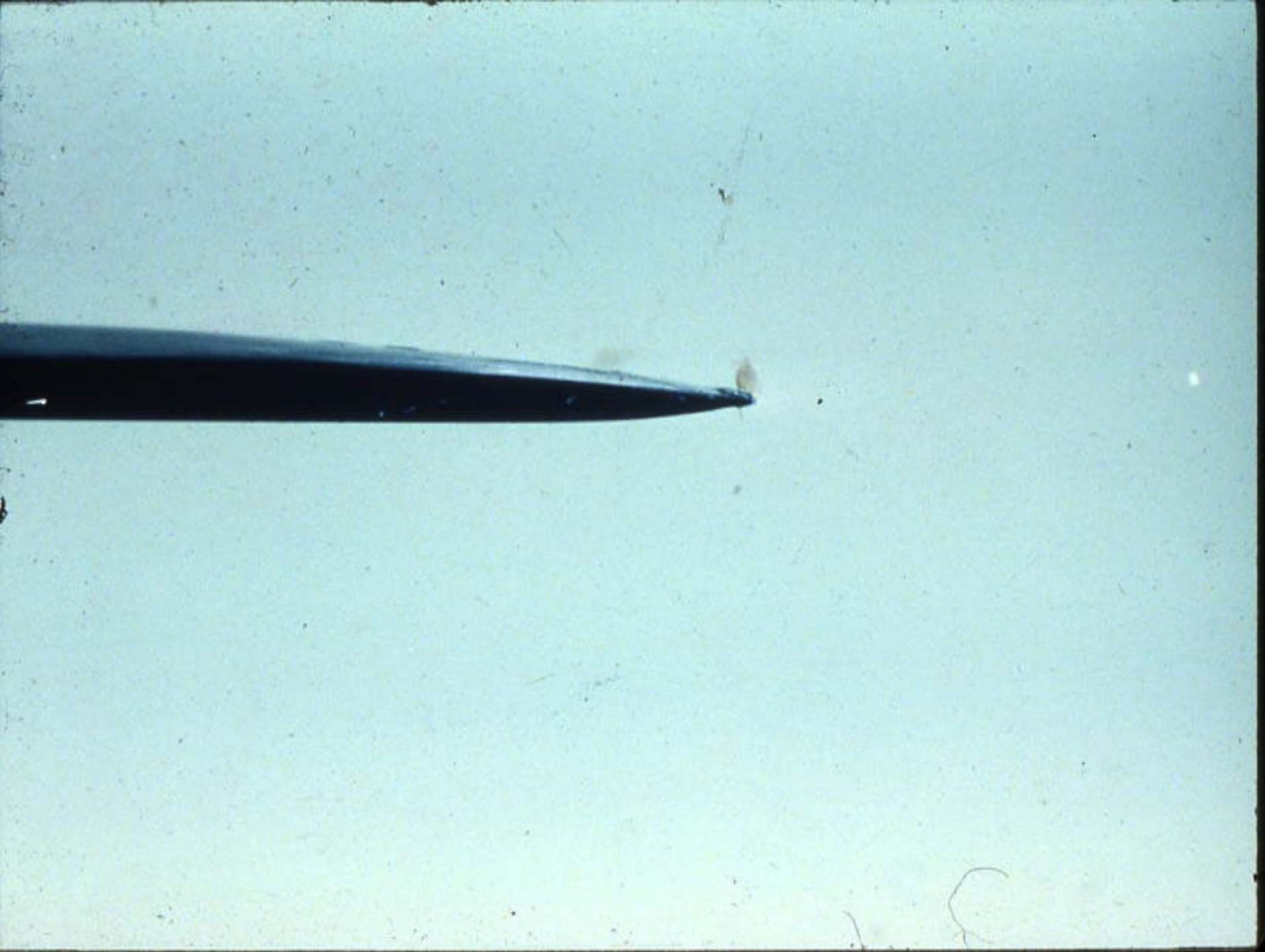
What is your diagnosis?





Scabies

- Infestation with a mite(*Sarcoptes scabiei*) that burrow into the stratum corneum
- Transmitted by physical contact
- Itchiness due to immune response to the mite
- Develop itchiness several weeks after being infested and will persist for weeks/months after treatment





Clinical features

- Itchy rash that spare the face in adult but could affect the face in infant
- Burrows especially in finger's and toe web
- Scabies nodules in genitalia and breast
- Excoriation marks without much inflammation









DDX

- Eczema
- Generalised pruritics of systemic disease
- Urticaria



Treatment

- Benzyl benzoate
- Gamma benzene hexachloride
- Permethrin
- Malathion
- Monosulfiram
- Treat the whole family
- 2x applications



Scabies pruritis after treatment

- Eurax
- Topical steroid
- Scabies nodule can be very resistant and need potent topical steroid

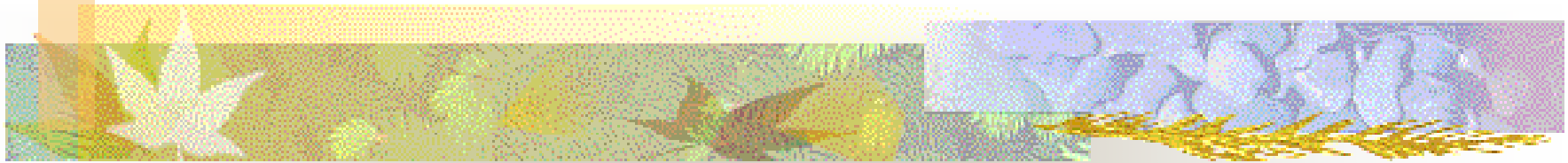


Case 4

- 65 yrs old elderly home resident with itchiness affecting her whole body including the face
- The symptom is persistent despite the use of topical steroid
- Clinical examination



What is the DDX?





Generalize pruritis of systemic disease

- Generalize itchiness affect the whole body without inflammatory dermatoses
- Need to ddx from scabies, mild asteatotic eczema, dermatographism.
- Difficult to manage



Causes

- Thyroid disease
- Parathyroid diseases
- Hepatic disease especially obstructive jaundice
- Lymphoma
- Uraemia
- Iron deficiency
- Myeloproliferative diseases



Investigation

- CBP
- RFT
- LFT
- Ca
- TSH
- ESR



Urticaria

- Acute: food and drug allergy, viral infection
- Chronic



Urticaria

- Idiopathic
- Infection (hepatitis, fungal infection, HP, parasite infestation)
- Systemic disease (SLE, autoimmune)
- Immunological (food, drug allergy)
- Physical
- Genetic







Chronic idiopathic urticaria

- 40% will resolve within 1 year
- associate with deep pressure urticaria
- about half of the patients have antibody against IgE or IgE receptor
- can associate with angiooedema and anaphylactic shock



Physical urticaria

- cholinergic
- deep pressure
- demographism
- cold
- heat
- aquagenic
- vibratory





Management of urticaria

- History and examination
- Physical challenge
- CBP, RFT, LFT, ESR, ANA, C3, C4, C1EI, Stool for ova
- Prick test, RAST
- Skin biopsy
- Oral challenge test



Case 5

- 66 yrs old man with several years history of rash affecting the extensor surfaces of his elbow and knee
- Mild itchy
- He has mark ‘danduff’ and nail changes affecting his fingers and toes
- Clinical examination



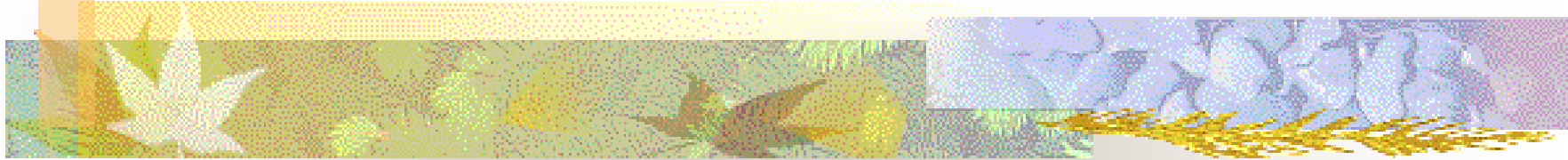


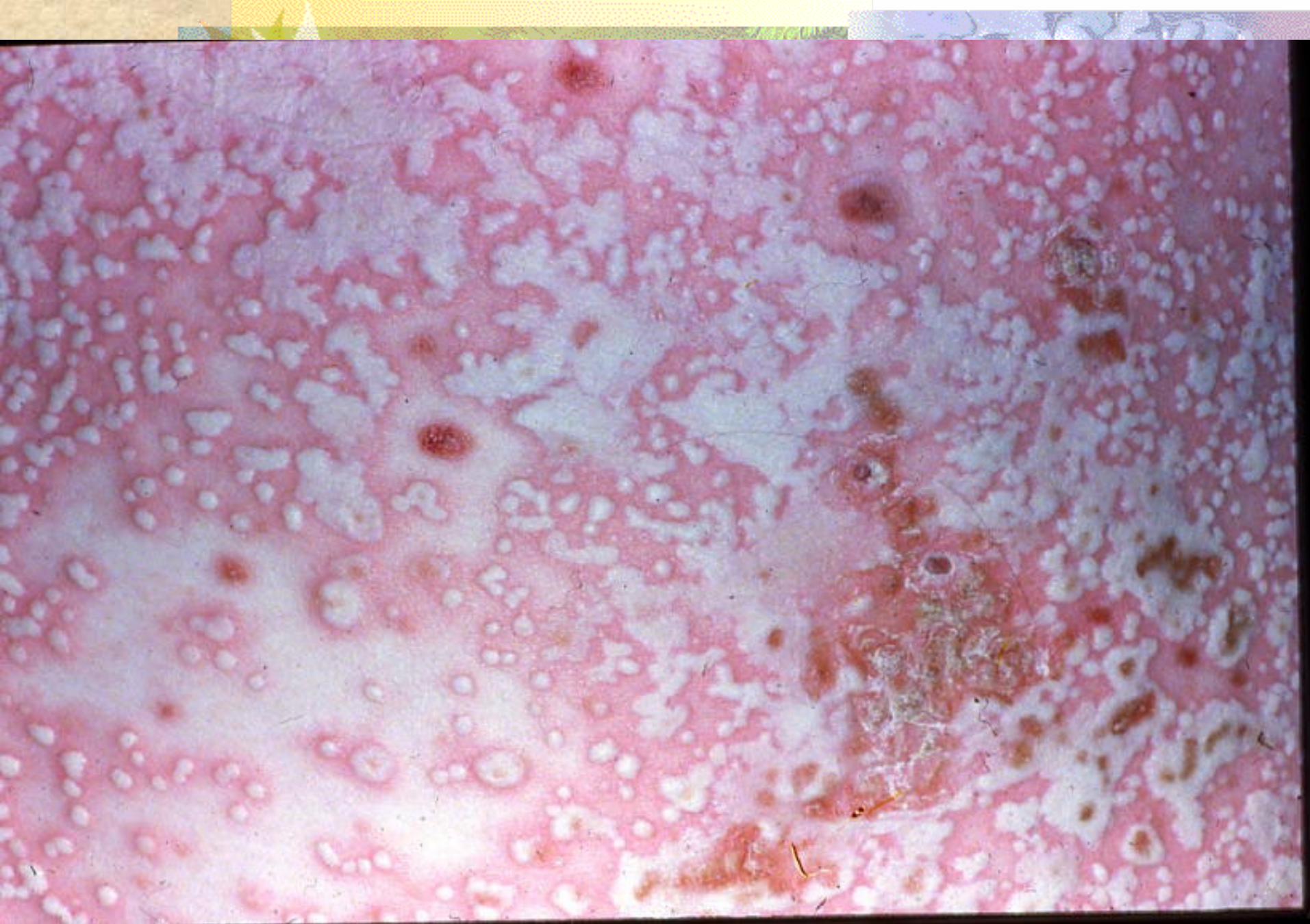


Case 6

- 37M, long history of skin problem with recent deterioration
- Seen by GP, given medication and injection
- Developed extensive rash with pustules

Clinical examination







These patients suffer from the same condition

- Diagnosis, DDX
- Investigation
- Management



Psoriasis

- Chronic inflammatory skin condition affecting 2-3% of the population
- 2 peak of onset: young and late middle age
- Earlier onset tends to be associated with worse prognosis
- genetic, drugs, infection (strept), stress





Clinical features

- chronic large plaque,
- guttate,
- pustular
- erythroderma
- Nail: pitting, onycholysis
- scalp, joints
- auspitz sign, koebner phenomenon



Treatment of psoriasis

- General counselling
- avoid aggravating factor
- eradicate infection



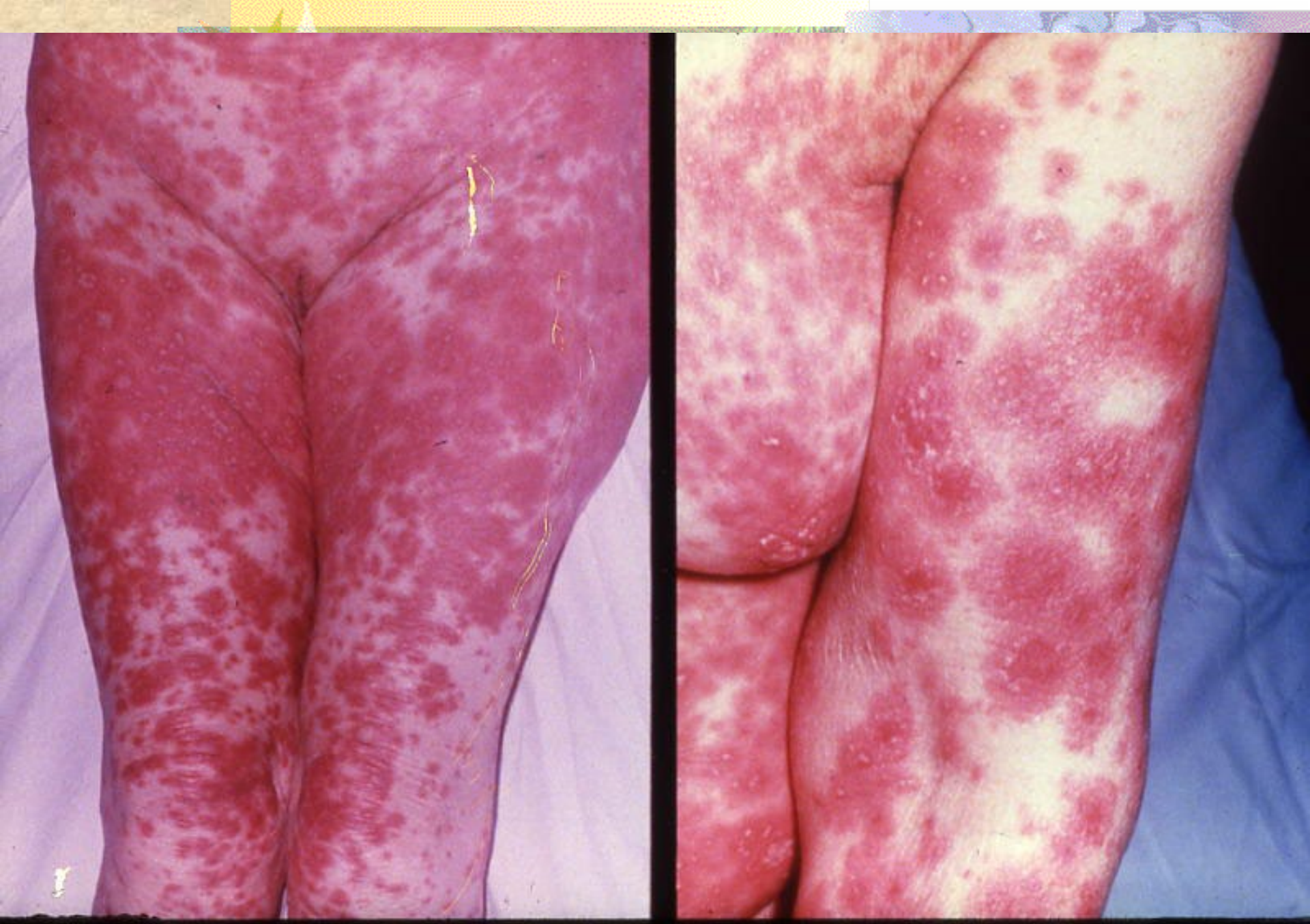
Management of case 5

- Dx clinically
- Management by the use of topical agents
 1. Diprosalic ointment in the morning
 2. Daivonex ointment at night
 3. Polytar shampoo



Differential diagnosis of case 6

- Pustular psoriasis
- Drug induced acute pustular eruption
- Staphylococcal scalded skin syndrome
- Subcorneal pustular dermatosis
- Dermatitis herpetiformis
- Pemphigus foliaceus







Investigation

- Skin swab and blood culture: pending
- CBP: WBC 15
- RFTLFT: N
- Ca: 2.0
- CXR: normal
- Drug history: was given IMI steroid PRN for several years for treatment of ‘psoriasis’
- Skin biopsy: pending



Management

- Bed rest
- Klacid 500mg BD
- Emulifying ointment as soap
- Aqueous cream LA BD
- IVI
- Careful fluid balance



Other investigations

- Skin swab and blood culture:
negative
- Skin biopsy: suggestive of
pustular psoriasis



Generalized pustular psoriasis(Von Zumbusch type)

- Sudden onset of sterile pustules with fever
- Erythema surrounds pustules
- Become extensive and confluent
- Nail involvement common
- Unusual to see other forms of psoriasis at the same time
- Often provoked by infection, drug(steroid withdraw, lithium, oral contraceptive)



Treatment

- Bed rest
- Treat or withdraw provoking factor
- Avoid irritant: tar, dithranol, phototherapy
- MTX, iso-acitretin, Cyclosporin



Patient progress

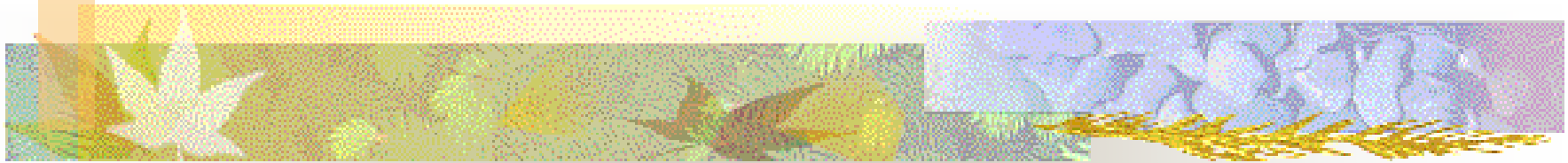
- Started on MTX initially 2.5mg as a testing dose and increase gradually to 15mg per week
- Careful monitor CBP,RFT,LFT
- Aim to gradual withdraw MTX
- Patient warned of the adverse effect and the need to have liver biopsy if he takes a total cumulative dose of 1.5g



1. The following statement is true about venous leg ulcer

1. It represents skin cancer
2. It is characterized by severe pain
3. It is usually associated with nail dystrophy
4. Compression dressing is useful
5. Absent of distal is hallmark of the disease

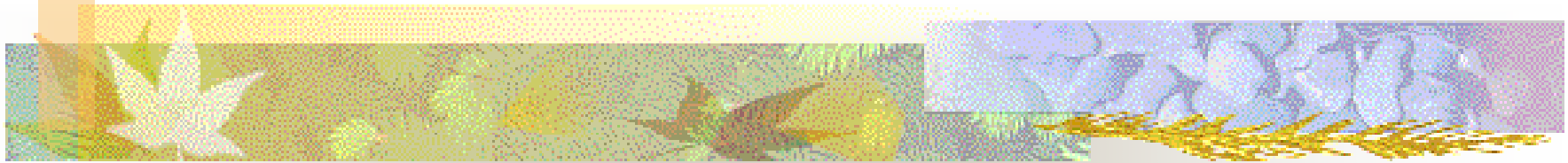
Answer: 4



2. Which of the following statement about psoriasis is correct

1. Well demarcated erythematous scaly plaques affecting the extensor surfaces
2. Do not involve the nails
3. Systemic steroid should be given
4. Do not involve the joints
5. It is a form of infection

Answer :1

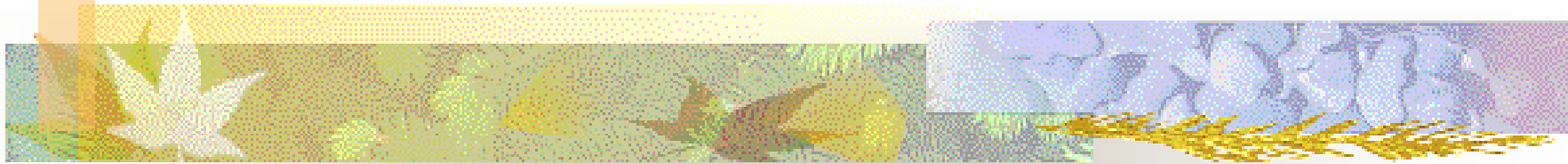




3. The following is correct about urticaria

1. It is not itchy
2. Uraemia is one of the causes
3. The rash lasted several days
4. Can be associated with systemic lupus
5. Is not associated with angioedema

Answer 4



Please tell us what you think of the MBBS
Whole Class Sessions.
Your opinions matter.



Login with your Portal ID

Look up (“the name of your teacher”) and answer ONE question.

[The Questionnaire will be released on the day of the teacher’s final WCS in the Block.]